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Dynamic Under Balanced Perforating Eliminates Near Wellbore Acid Stimulation in Low-Pressure Weber Formation

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Abstract

Completions in the Weber formation have historically been perforated over balanced and then acid stimulated. Flow rates before acidizing were virtually 0 Mcf/D. Acid was an important step in the completion process. The Weber gas reservoir has low permeability (0.5 - 1.5 mD) and is now low pressure (2800 to 2300 psi at 14,000 ft). Typical flowing rates for these wells range from 1 to 5 MMcf/D. A new perforating technique utilizing the dynamic under balanced method was implemented in two wells, Brady 38W and Brady 56W. The wells were perforated with tubing-conveyed guns with initial *over balances* of 3250 and 3750 psi. The dynamic under balanced method creates an instantaneous under balance even though the annular space between the gun and the casing is initially over balanced. This perforating method also leaves significantly less perforation "skin." The proof is in the success of the two wells, which have rates equal to or in excess of the existing wells and yet required no acid stimulation. From historical comparisons of 18 prior completions, these wells should have had nearly zero flow without acid when perforating using an initial over balance of up to 3750 psi in the annulus. This paper discusses the new dynamic under balanced perforating design and its application to low-permeability, under-pressured reservoirs.

Background

The Brady Field was discovered in 1973 with the completion of the Brady 1W well in the sour Weber formation at a depth of 14,000'. The original reservoir pressure was 5700 psi or 0.41 psi/ft. 14 additional wells were drilled from 1973 until 1975 and completed in the same Weber formation. A plant was constructed to treat and recycle the gas through the formation. One additional well was drilled in 1982 and then another in 1985. The field extracted liquids from 1973 until 1998 when the recycle process was discontinued and the field went into blowdown phase. The Weber formation consists of approximately 600' of interbedded sandstone with multiple shale and dolomite stringers. The completion practices for this formation typically consisted of selectively perforating 1 SPF over several intervals with casing guns under a full column of fluid. After perforating, the tubing and packer were run and the wellbore fluid was displaced with nitrogen. A flow test was then performed. The usual poor performance of the wells required an HCL/HF acid treatment that generally produced much more favorable results. Three of the 18 wells had to be hydraulically fractured to improve performance.

In 2002, a recompletion in the Brady 38W to the upper section of a producing Weber well was attempted using dynamic under balanced perforating. This process was chosen due to the complexity of the wellbore. Figure 1 illustrates how the presence of some old squeeze perfs above the zone of interest complicated the ability to do a remedial acid treatment. The existence of squeeze perfs also created complexity when considering the need for a possible fracture stimulation if perforating alone did not create the desired results. It was determined that dynamic under balanced perforating provided the best possible chance for a non-stimulated completion. The greatest challenge in applying this technique was recognizing and accepting that the initial static conditions just before perforating would actually be 3250 psi over balanced. Without creating a dynamic condition, this well would surely need additional stimulation.

Later in the same year, Anadarko decided to drill a new well in the Weber formation. This was the first new Weber well drilled in over 17 years. This new well provided the option to choose from any perforating and stimulation combination desired. The results of the dynamic under balanced perforating technique on the Brady 38W were strong enough that the same perforating technique was chosen for this newly drilled well, the Brady 56W. In this case, the pre-perforating static conditions would be even higher than the Brady 38W at 3750 psi over balanced. Again, based on history, this would have certainly been a candidate for additional stimulation unless we were able to create a dynamic under balanced condition. The results of this completion were likewise very successful and created enough interest to warrant investigating other applications of this perforating technique in other fields around the world.

Pre-job Nodal analysis

As a part of the process of determining what to expect from the recompletion process, Nodal analysis was performed to simulate various possible conditions. Matches were made from early well performance data and an estimate was made for what rate a "zero-skin" perforating condition would yield. Figure 2 is a representation of the pre-job Nodal summary for the Brady 38W. At the current bottom hole pressure estimate, this well should have been capable of producing approximately 3.85MMcfd with no perforation skin. Of course without acid, the skin historically would have exceeded 20 and the rate would have been minimal.

Under balanced perforating

King, et al[1] published field test data for oil and gas wells that were completed with tubing conveyed perforating (TCP). The wells were completed with a range of under balances (the condition where well bore pressure is less than reservoir pressure). All wells were acid stimulated to cleanup perforating damage. The productivity index, PI, was measured after perforating and after acid stimulation. If the PI after stimulation was less than a 10% increase from perforating alone, then King suggested that the level of applied under balance was sufficient and near well bore acid stimulation was not required. Tariq[2], calculated a statistical best fit between the two data sets (acid required and acid not required, see Fig. 9 of Reference 2). Equation 1 gives the minimum required under balance to eliminate acid stimulation.

$$\text{Minimum under balance, delta P} = 3000/K^{0.4} \qquad (1)$$

For the nominal 0.5 – 1.5md Weber formation, the above equation gives the range of maximum-to-minimum under balance as 3960 to 2550 psi, considerably in excess of its 2800 to 2300 psi reservoir pressure. The implication is that even if the well was perforated dry, the maximum under balance would be 2800 psi or less in the more depleted sections, and thus, near well bore acid stimulation would be required for the well to flow at its full potential. Additionally, King never implied that if the minimum under balance were used, all perforations would be clean. By clean, we mean 100% effective shot density and no perforation crushed damage zone.

Using the minimum under balance equation from Behrmann[3], and assuming 8% porosity, 6400 psi unconfined compressive strength (UCS) and permeability of 0.5 – 1.5md, we calculate a maximum-to-minimum under balance to obtain clean perforations of 6390 to 4430 psi. This requires an under balanced condition that is substantially in excess of the reservoir pressure.

Recent[4] internal Schlumberger perforating tests in 1-2md gas saturated rock at residual brine saturation gave a single shot core flow efficiency (CFE) of 0.64 for a 2000 psi under balance. CFE is the experimental PI divided by the theoretical PI for an equivalent sized undamaged perforation. The perforation cavity was completely filled with sand and perforation liner debris as shown in the CT-scan of Figure 3. A dynamic under balance test resulted in a CFE of one. The result was a large diameter cavity and a perforation void of sand debris for about 75% of the perforated depth as shown in Figure 4. This test provided encouragement that the application of dynamic under balance may provide clean perforations in a low permeability gas reservoir.

Dynamic under balanced perforating

A dynamic under balance is the actual under balance obtained during approximately the first 100 milliseconds after the initiation of a perforating event. This dynamic under balance is a function of: 1) the initial well bore and reservoir pressures, 2) the perforating gun, 3) charge design, 4) well bore geometries, and 5) the reservoir permeability. A dynamic under balance can be obtained independent of whether or not the initial static conditions are under balanced, over balanced, or balanced[5-6]. To obtain a dynamic under balance, the perforating system and process is designed to create an instantaneous under balance during perforating. This rapid dynamic under balance is the result of the high-pressure wellbore fluid filling up the empty guns and tubing and thus lowering the wellbore pressure. This under balance must be created immediately after charge detonation (less than 0.1 sec) to effectively remove the crushed zone. Therefore, the perforating string must be designed accordingly. The surge flow into the well bore and into the perforating gun system removes the crushed damage zone particles from the perforating cavities. To prevent the well bore from returning to an over balanced state before the well bore and pore pressure are equalized, the perforating should be done under a packer with the tubing closed[7].

Examples of actual dynamic under balances for the tests shown in Figures 3 and 4 are shown in Figures 5 and 6. Note that for the conventional 2000 psi static under balance test, the actual dynamic under balance was less than ~200 psi and did not occur until about 0.120 seconds after the charge was fired. On the other hand, the optimized dynamic under balance of test two created a dynamic under balance of 2000 psi within 0.025 seconds after the charge was fired in spite of starting with a 1000 psi over balance.

Bottom Hole Assembly (BHA)

The Brady wells were TCP conveyed permanent completions. Because of the low permeability and the low pressure gas reservoir, the completion string and perforating processes were designed to allow immediate flow and cleanup following perforating. This was done by minimizing the amount of liquid that was in the tubing that the gas would have to lift during the initial flow cleanup period. The bottom hole pressure (BHP) with the well full of completion fluid is 6050 psi. In order to have the tubing hydrostatic head be much less than the reservoir pressure, the BHA and the annular space under the packer (rathole) needed to be isolated from the tubing, but at the same time open to the tubing immediately upon perforating. To accomplish this, a special production valve was used that provides isolation between the tubing, the guns, and the annular space below the packer and it opens when the firing head is initiated. Thus, there is an almost instantaneous flow from the rathole and reservoir into the tubing when the guns are fired. A well bore diagram is provided in Figure 7 to further illustrate the BHA.

A generic BHA, from bottom up is: 2-7/8 inch high shot density (HSD) perforating guns designed to create a dynamic under balance, a fast acting production valve, firing heads (drop bar and hydraulic – one redundant), a sliding sleeve, and a packer. An additional consideration in the design requires a calculation of what the final wellbore pressure would be if the production valve does not open. Although a dynamic under balance is created, if the combination of rathole volume, wellbore pressure, gun internal volume, and explosive mass are not correct, the wellbore pressure could go from over balanced to under balanced and back to over balanced. This could result in damaged perforations. There was sufficient gun volume on the Brady 38W that this was not an issue. However, the 3000 ft. rathole on the Brady 56W required additional empty TCP guns.

Operational Procedure

With the well full of completion fluid, the TCP assembly was run in the hole with the sliding sleeve open. Once on depth, the packer was set and the sliding sleeve closed, trapping 6050 psi in the annular space under the packer (rathole). The tubing was swabbed to a liquid level of about 1000 ft. above the packer or 12,000' below the surface. The initial static conditions in the wellbore at this point are over balanced and are illustrated in Figure 8. The surface equipment was installed, the tubing was left open, and the drop bar was released. After a specified time, the delayed firing head detonated. When the firing head detonated, instantly the production valve opened and a dynamic under balance was generated. With 2800 to 2300 psi in the reservoir, 6050 psi in the annular space under the packer, and 450 psi in the tubing, all fluids in the system flowed immediately to surface.

In the event that the drop bar firing head had failed to function, the backup hydraulic head would have been employed by pressuring up the tubing with gas. The gas would then have been bled off during the time delay of the firing head to evacuate the tubing of any backpressure.

Although in this case the guns were left hanging, a mechanical gun drop sub could have been used and the guns dropped in the rathole after the well cleaned up.

Brady 38W results. The pre-job Nodal analysis indicated an expected flow rate of 3.85MMcfd if the perforation skin was zero. Historically, much lower flow rates and much higher perforation skins (in excess of 20) were encountered when perforating over balanced and without acid treatment. The well achieved a sustained flow rate of 5.2 MMcfd within hours of perforating. It is estimated that the actual skin through the perfs is –1.17 due to the enhancement created by the dynamic under balanced technique.

Brady 56W issues. The distance from the packer to the well bottom for Brady 56W was about 3000 ft. The distance to the top perforation under the packer was 2000 ft. As already mentioned, this large rathole volume required additional lengths of empty guns to assure that the rathole stayed under balanced or balanced after the creation of the dynamic under balance. The well experienced some difficulty in unloading all of the annular fluid and ultimately it was determined that the reservoir had a lower than expected bottom hole pressure (2300 psi versus 2800 psi). Nevertheless, after sufficient time for cleanup, the well performed at rates above that expected for a well with no perforation skin. The pre-job Nodal analysis was modified to reflect the lower BHP. The new expected rate was 3.0MMcfd. The well actually flowed at a sustained rate of 4.2 MMcfd after clean-up. This rate can be achieved in the Nodal analysis by inputting a perforation skin of -1.2.

Conclusions

- The new dynamic under balanced perforating technique was successfully applied to the low permeability, low pressure Weber gas reservoir resulting in calculated perforation skin values of –1.2. This is a significant improvement over historical values of over 20 when perforated balanced.
- The application of dynamic under balanced perforating eliminated the historical need for expensive acid treatments or hydraulic fracture stimulations. Despite initial static over balances of 3250 and 3750 psi, the wells flowed naturally after perforating.
- The application of dynamic under balanced perforating resulted in quicker gas sales than the conventional method and eliminated other operational steps that are expensive and safety sensitive in this sour gas environment.
- The conventional under balance requirements do not appear to apply for the dynamic under balance system and in fact over estimate the required dynamic under balance needed for optimal results. Clearly, additional physics need to be considered to understand perforation cleanup under a dynamic event.

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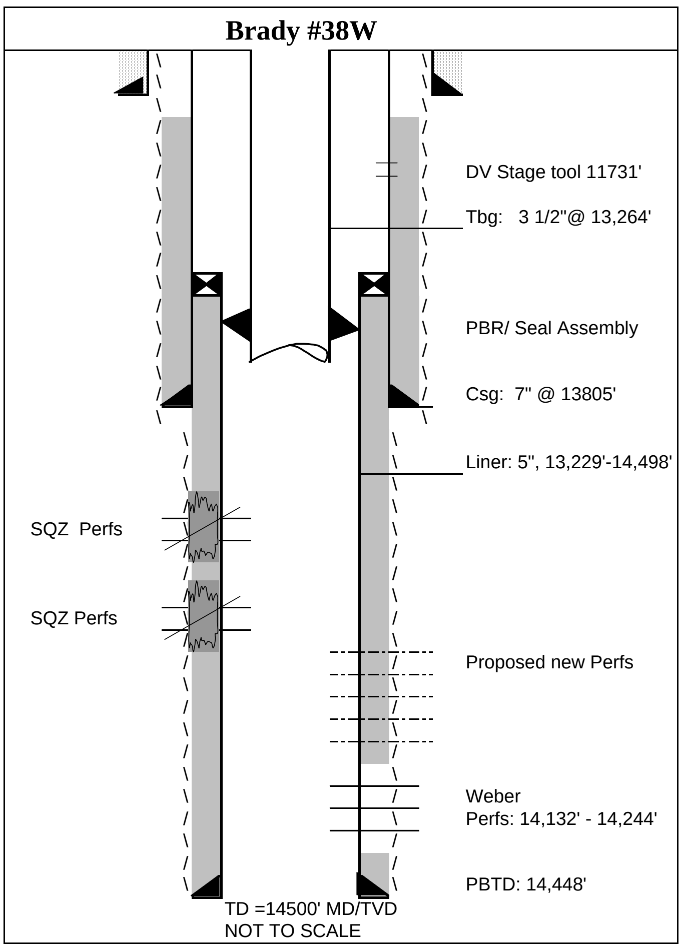

Figure 1 Wellbore diagram of the Brady 38W illustrating the difficulty of having to acidize or hydraulically fracture after perforating.

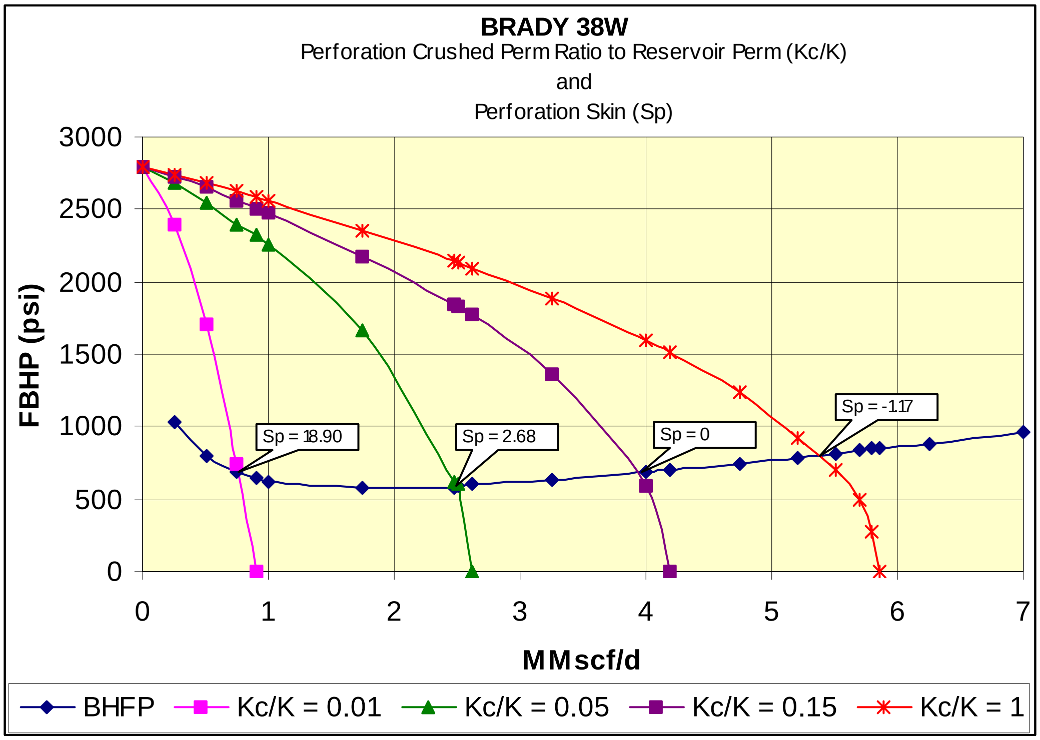

Figure 2 shows the effect that improving perforation skin (Sp) has on flow rate. Historical wells had an Sp in excess of 20 and very low flow rates. The dynamic underbalanced perforating technique yielded an Sp of -1.17 and a flow rate of 5.2MMcfd.

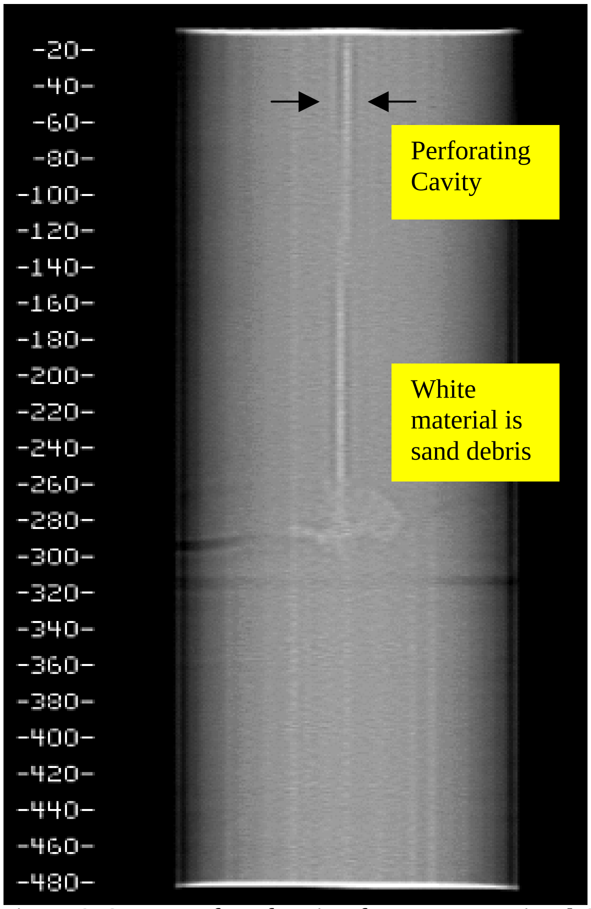

Figure 3 CT-scan of perforation from a conventional 2000 psi under balance test.

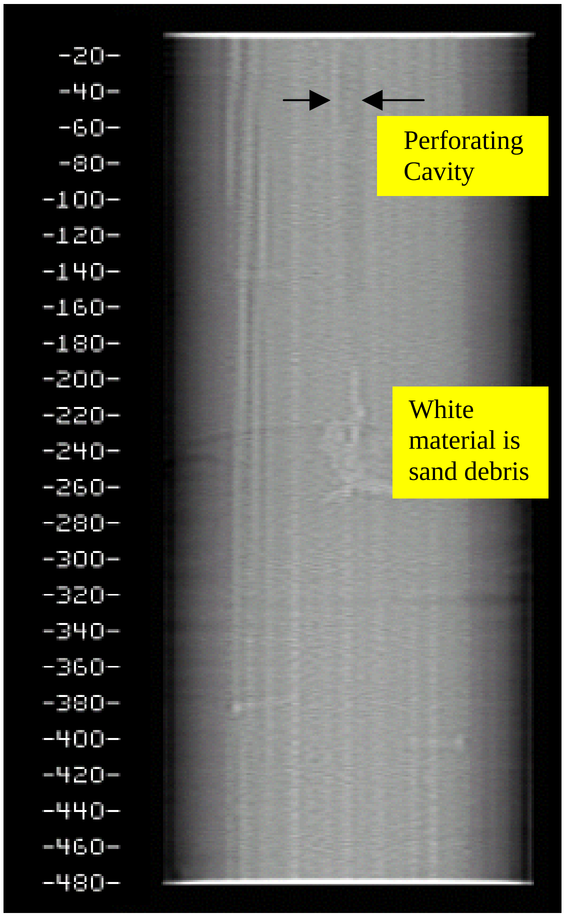

Figure 4 CT-scan of perforation from a dynamic under balance with an initial 1000 psi over balance.

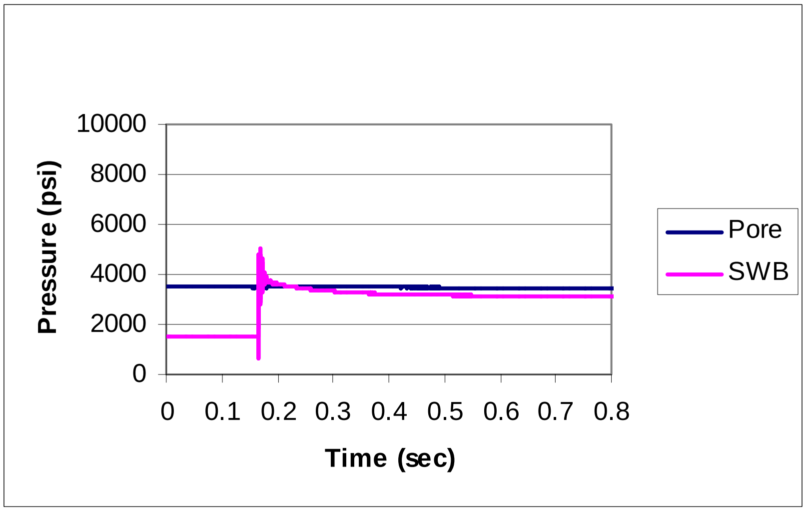

Figure 5 Pressure traces from a conventional 2000 psi initial under balance. SWB = simulated wellbore. Test 1 from Fig. 3.

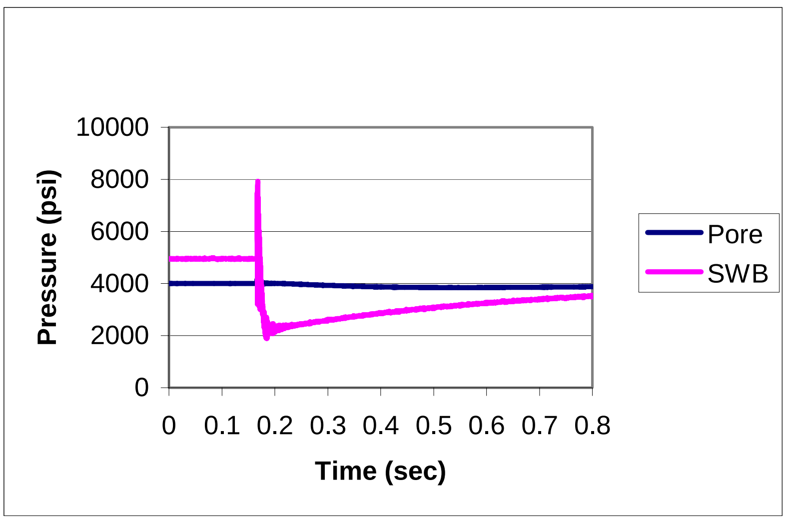

Figure 6 Pressure traces from an optimized dynamic under balance starting from a 1000 psi static over balance, Test 2 from Fig. 4.

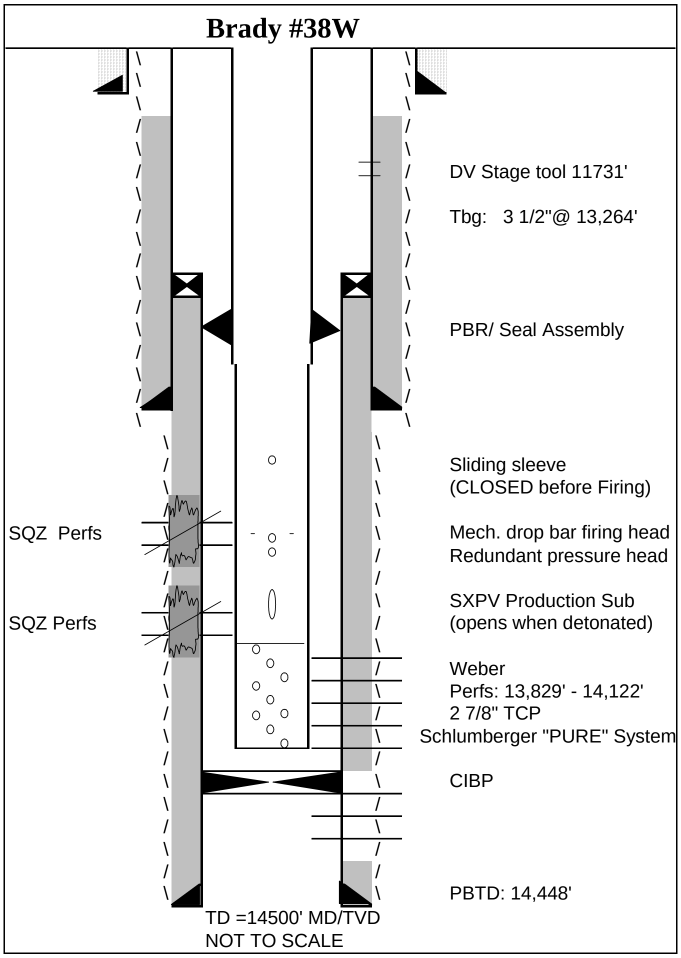

Figure 7 Wellbore diagram illustrating the BHA of the "PURE" dynamic under balanced perforating system used on the Brady 38W.

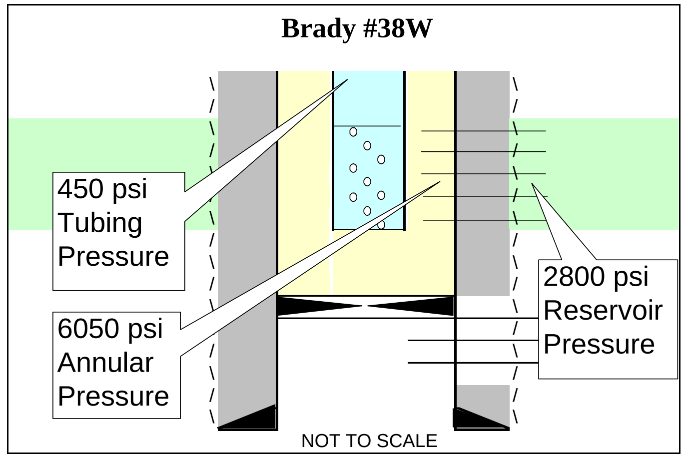

Figure 8 Initial static conditions just before perforating the Brady 38W.